

```
package ques/FinancialForecasting;

public class FinancialForecast {
    // Recursive Method
    public static double predictFutureValue(double currentValue, double growthRate, int years) {

        // Base Case
        if (years == 0) {
            return currentValue;
        }

        // Recursive Call
        return predictFutureValue(currentValue * (1 + growthRate), growthRate, years - 1);
    }
}
```

```
package ques7FinancialForecasting;

public class ForecastTest {
    public static void main(String[] args) {

        double currentValue = 10000;
        double growthRate = 0.10; // 10%
        int years = 5;

        double futureValue = FinancialForecast.predictFutureValue(currentValue, growthRate, years);

        System.out.println("Current Value : " + currentValue);
        System.out.println("Growth Rate : " + (growthRate * 100) + "%");
        System.out.println("Years : " + years);
        System.out.println("Future Value : " + futureValue);
    }
}
```

Current Value : 10000.0

Growth Rate : 10.0%

Years : 5

Future Value : 16105.100000000008